

BABU BANARASI DAS
UNIVERSITY

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SCHOOL OF COMPUTER APPLICATIONS

NO SQL PROJECT ON
Student Management System Using MongoDB

Submitted to:

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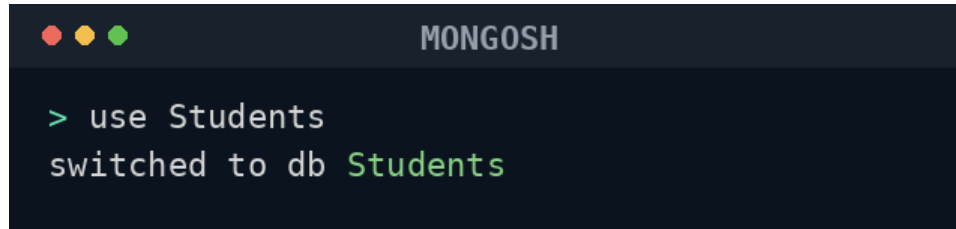
Section: BCADS23

1. CREATE DATABASE

MongoDB Query:

use Students

Explanation: Creates or switches to the Student database.



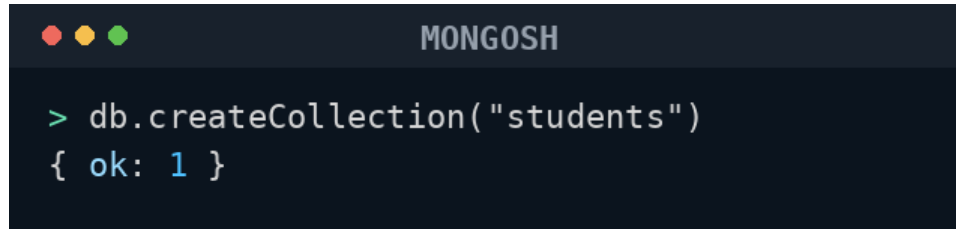
```
MONGOSH  
  
> use Students  
switched to db Students
```

2. CREATE COLLECTION

MongoDB Query:

db.createCollection("students")

Explanation: Creates a collection named students.

A terminal window with a dark background and a title bar containing three colored window control buttons (red, yellow, green) and the text "MONGOSH". The terminal shows a command prompt ">" followed by the command "db.createCollection('students')". Below the command, the output is displayed as "{ ok: 1 }".

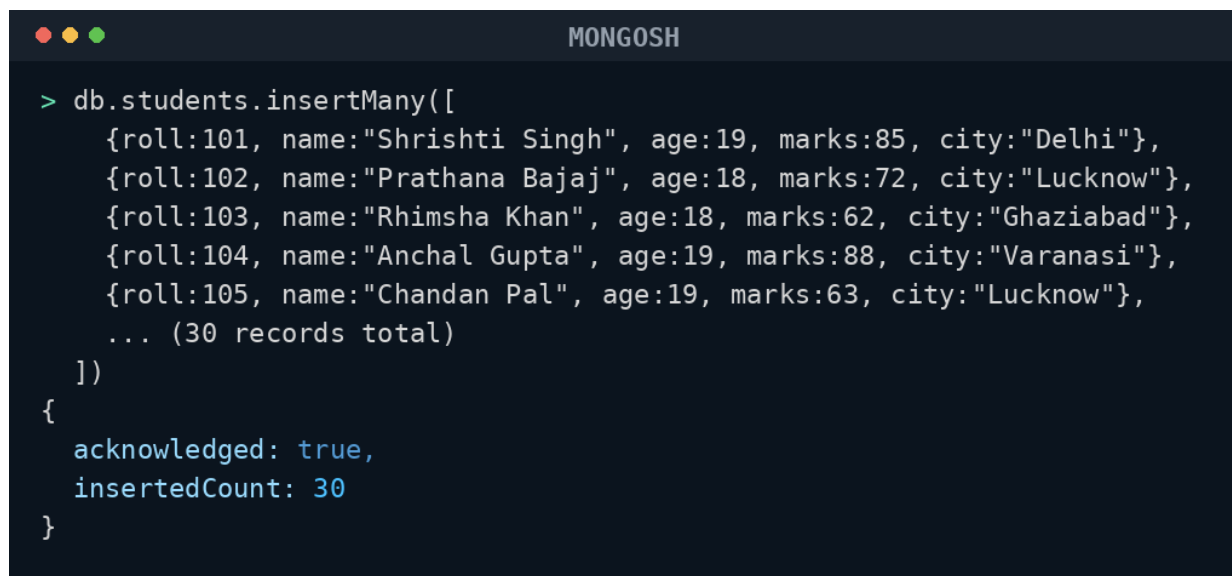
```
> db.createCollection("students")
{ ok: 1 }
```

3. INSERT 30 STUDENTS

MongoDB Query:

```
db.students.insertMany([
  {roll:101, name:"Shrishti Singh", age:19, marks:85, city:"Delhi"},
  {roll:102, name:"Prathana Bajaj", age:18, marks:72, city:"Lucknow"},
  {roll:103, name:"Rhimsha Khan", age:18, marks:62, city:"Ghaziabad"},
  {roll:104, name:"Anchal Gupta", age:19, marks:88, city:"Varanasi"},
  {roll:105, name:"Chandan Pal", age:19, marks:63, city:"Lucknow"},
  {roll:106, name:"Ishika Shrivastava", age:19, marks:83, city:"Prayagraj"},
  {roll:107, name:"Kritika Verma", age:18, marks:65, city:"Ghaziabad"},
  {roll:108, name:"Saksham Rao", age:19, marks:71, city:"Ghaziabad"},
  {roll:109, name:"Devyani Joshi", age:19, marks:93, city:"Ghaziabad"},
  {roll:110, name:"Aryan Malhotra", age:19, marks:61, city:"Varanasi"},
  {roll:111, name:"Nikhil Ali", age:20, marks:54, city:"Ghaziabad"},
  {roll:112, name:"Rohan Tripathi", age:20, marks:95, city:"Prayagraj"},
  {roll:113, name:"Tanisha Kapoor", age:19, marks:57, city:"Delhi"},
  {roll:114, name:"Faizan Ahmed", age:19, marks:85, city:"Prayagraj"},
  {roll:115, name:"Vanshika Rawat", age:20, marks:60, city:"Lucknow"},
  {roll:116, name:"Lakshay Sethi", age:18, marks:61, city:"Noida"},
  {roll:117, name:"Muskan Jaiswal", age:19, marks:84, city:"Varanasi"},
  {roll:118, name:"Harshvardhan Yadav", age:20, marks:90, city:"Ghaziabad"},
  {roll:119, name:"Gungun Mishra", age:20, marks:94, city:"Lucknow"},
  {roll:120, name:"Sarthak Bhatt", age:20, marks:61, city:"Prayagraj"},
  {roll:121, name:"Alisha Khan", age:20, marks:95, city:"Noida"},
  {roll:122, name:"Divyanshu Pandey", age:19, marks:93, city:"Lucknow"},
  {roll:123, name:"Kashish Arora", age:20, marks:94, city:"Delhi"},
  {roll:124, name:"Rajat Chauhan", age:19, marks:81, city:"Delhi"},
  {roll:125, name:"Shivangi Dubey", age:20, marks:61, city:"Ghaziabad"},
  {roll:126, name:"Yashwant Rana", age:18, marks:60, city:"Delhi"},
  {roll:127, name:"Bhavya Saxena", age:19, marks:73, city:"Ghaziabad"},
  {roll:128, name:"Nitya Agarwal", age:18, marks:63, city:"Delhi"},
  {roll:129, name:"Om Prakash", age:19, marks:95, city:"Ghaziabad"},
  {roll:130, name:"Tanmay Srivastava", age:19, marks:72, city:"Varanasi"}
])
```

Explanation: Inserts multiple student documents into the collection.



```
> db.students.insertMany([
  {roll:101, name:"Shrishti Singh", age:19, marks:85, city:"Delhi"},
  {roll:102, name:"Prathana Bajaj", age:18, marks:72, city:"Lucknow"},
  {roll:103, name:"Rhimsha Khan", age:18, marks:62, city:"Ghaziabad"},
  {roll:104, name:"Anchal Gupta", age:19, marks:88, city:"Varanasi"},
  {roll:105, name:"Chandan Pal", age:19, marks:63, city:"Lucknow"},
  ... (30 records total)
])
{
  acknowledged: true,
  insertedCount: 30
}
```

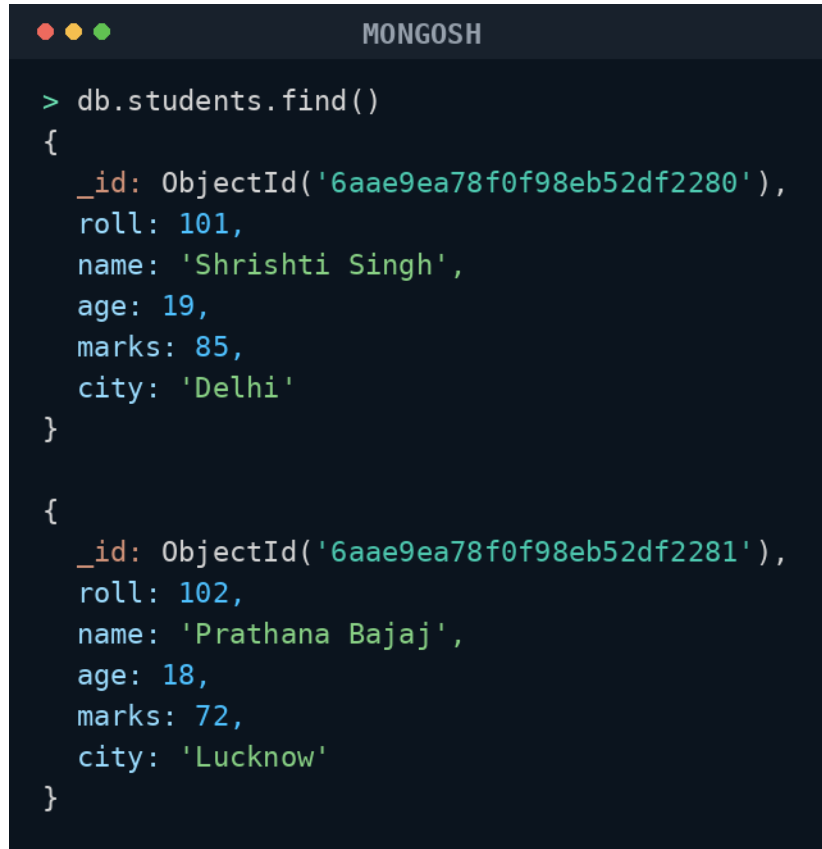
All 30 student records were inserted successfully into the 'students' collection.

4. DISPLAY ALL STUDENTS

MongoDB Query:

`db.students.find()`

Explanation: Retrieves documents from the students collection.



```
> db.students.find()
{
  _id: ObjectId('6aae9ea78f0f98eb52df2280'),
  roll: 101,
  name: 'Shrishti Singh',
  age: 19,
  marks: 85,
  city: 'Delhi'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df2281'),
  roll: 102,
  name: 'Prathana Bajaj',
  age: 18,
  marks: 72,
  city: 'Lucknow'
}
```

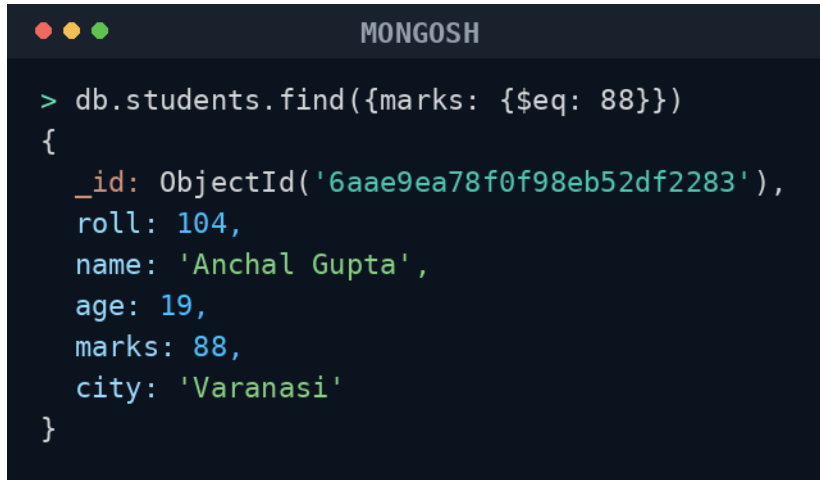
Showing the first 2 of 30 documents returned by this query.

5. EQUAL TO — \$eq

MongoDB Query:

```
db.students.find({marks: {$eq: 88}})
```

Explanation: Finds students whose marks are exactly 88.



```
MONGOSH
> db.students.find({marks: {$eq: 88}})
{
  _id: ObjectId('6aae9ea78f0f98eb52df2283'),
  roll: 104,
  name: 'Anchal Gupta',
  age: 19,
  marks: 88,
  city: 'Varanasi'
}
```

6. GREATER THAN — \$gt

MongoDB Query:

```
db.students.find({marks: {$gt: 85}})
```

Explanation: Finds students whose marks are greater than 85.

```
MONGOSH

> db.students.find({marks: {$gt: 85}})
{
  _id: ObjectId('6aae9ea78f0f98eb52df2283'),
  roll: 104,
  name: 'Anchal Gupta',
  age: 19,
  marks: 88,
  city: 'Varanasi'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df2288'),
  roll: 109,
  name: 'Devyani Joshi',
  age: 19,
  marks: 93,
  city: 'Ghaziabad'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df228b'),
  roll: 112,
  name: 'Rohan Tripathi',
  age: 20,
  marks: 95,
  city: 'Prayagraj'
}
```

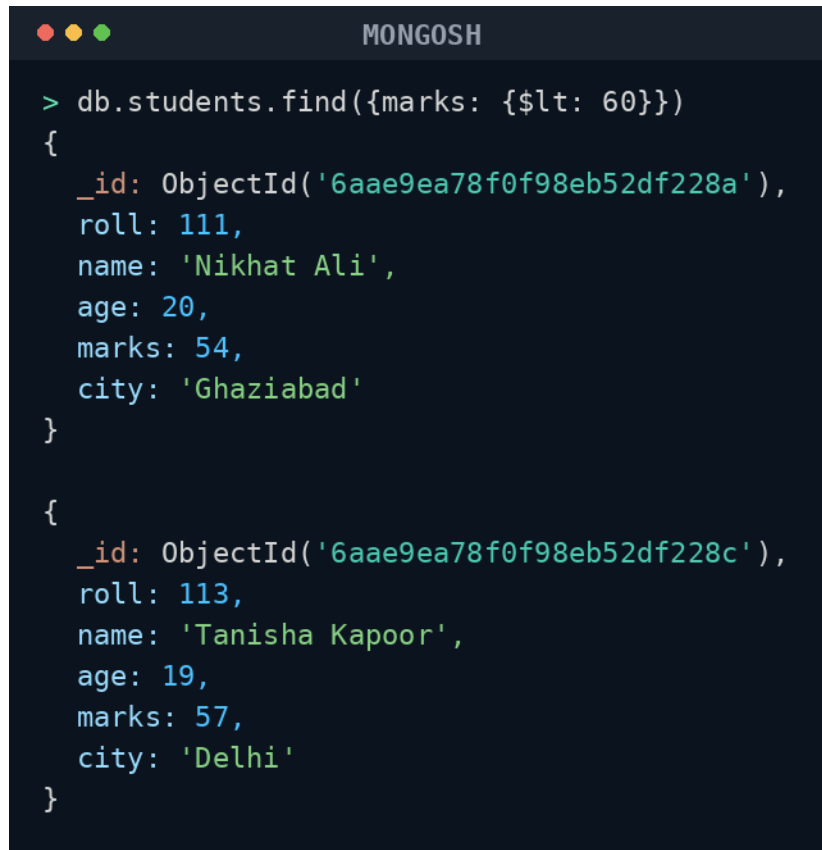
Showing 3 of 9 matching documents.

7. LESS THAN — \$lt

MongoDB Query:

`db.students.find({marks: {$lt: 60}})`

Explanation: Finds students whose marks are less than 60.

A screenshot of a terminal window titled "MONGOSH". The terminal shows a MongoDB query being executed: `> db.students.find({marks: {$lt: 60}})`. The output consists of two JSON documents. The first document represents a student with `_id: ObjectId('6aae9ea78f0f98eb52df228a')`, `roll: 111`, `name: 'Nikhat Ali'`, `age: 20`, `marks: 54`, and `city: 'Ghaziabad'`. The second document represents a student with `_id: ObjectId('6aae9ea78f0f98eb52df228c')`, `roll: 113`, `name: 'Tanisha Kapoor'`, `age: 19`, `marks: 57`, and `city: 'Delhi'`.

```
> db.students.find({marks: {$lt: 60}})
{
  _id: ObjectId('6aae9ea78f0f98eb52df228a'),
  roll: 111,
  name: 'Nikhat Ali',
  age: 20,
  marks: 54,
  city: 'Ghaziabad'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df228c'),
  roll: 113,
  name: 'Tanisha Kapoor',
  age: 19,
  marks: 57,
  city: 'Delhi'
}
```

8. GREATER THAN OR EQUAL TO — \$gte

MongoDB Query:

```
db.students.find({marks: {$gte: 90}})
```

Explanation: Finds students whose marks are 90 or above.

A screenshot of a terminal window titled "MONGOSH". The terminal shows a MongoDB query being executed: `> db.students.find({marks: {$gte: 90}})`. The output displays three JSON documents. The first document is for a student with roll number 109, name 'Devyani Joshi', age 19, marks 93, and city 'Ghaziabad'. The second document is for a student with roll number 112, name 'Rohan Tripathi', age 20, marks 95, and city 'Prayagraj'. The third document is for a student with roll number 118, name 'Harshvardhan Yadav', age 20, marks 90, and city 'Ghaziabad'. The documents are formatted with syntax highlighting: ObjectId is orange, roll is blue, name is green, age is blue, marks is blue, and city is green.

```
> db.students.find({marks: {$gte: 90}})
{
  _id: ObjectId('6aae9ea78f0f98eb52df2288'),
  roll: 109,
  name: 'Devyani Joshi',
  age: 19,
  marks: 93,
  city: 'Ghaziabad'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df228b'),
  roll: 112,
  name: 'Rohan Tripathi',
  age: 20,
  marks: 95,
  city: 'Prayagraj'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df2291'),
  roll: 118,
  name: 'Harshvardhan Yadav',
  age: 20,
  marks: 90,
  city: 'Ghaziabad'
}
```

Showing 3 of 8 matching documents.

9. LESS THAN OR EQUAL TO — \$lte

MongoDB Query:

```
db.students.find({marks: {$lte: 70}})
```

Explanation: Finds students whose marks are 70 or below.

```
MONGOSH

> db.students.find({marks: {$lte: 70}})
{
  _id: ObjectId('6aae9ea78f0f98eb52df2282'),
  roll: 103,
  name: 'Rhimsha Khan',
  age: 18,
  marks: 62,
  city: 'Ghaziabad'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df2284'),
  roll: 105,
  name: 'Chandan Pal',
  age: 19,
  marks: 63,
  city: 'Lucknow'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df2286'),
  roll: 107,
  name: 'Kritika Verma',
  age: 18,
  marks: 65,
  city: 'Ghaziabad'
}
```

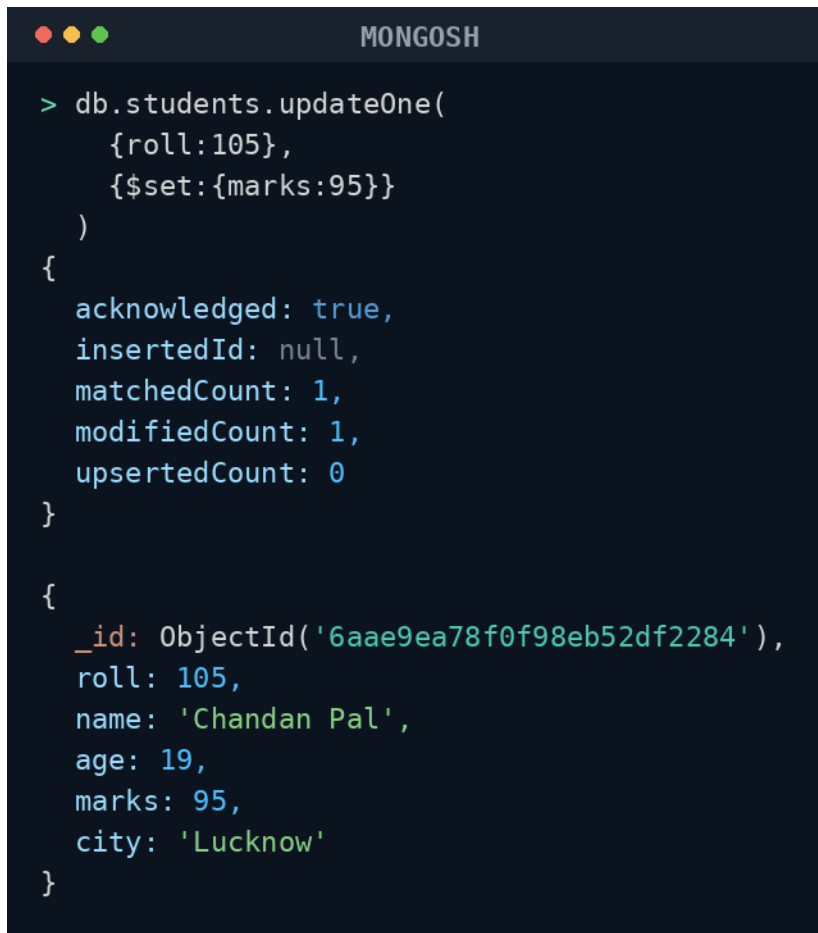
Showing 3 of 12 matching documents.

10. UPDATE OPERATION

MongoDB Query:

```
db.students.updateOne(  
  {roll:105},  
  {$set:{marks:95}}  
)
```

Explanation: Updates the marks to 95 of the student with roll number 105.



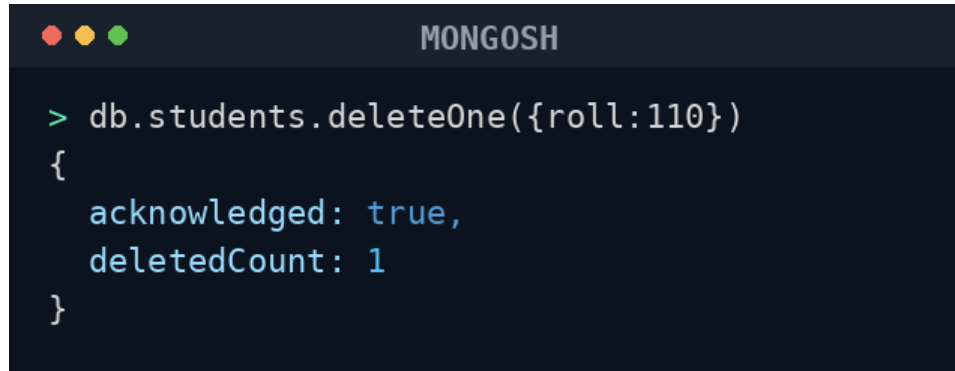
```
MONGOSH  
  
> db.students.updateOne(  
  {roll:105},  
  {$set:{marks:95}}  
)  
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}  
  
{  
  _id: ObjectId('6aae9ea78f0f98eb52df2284'),  
  roll: 105,  
  name: 'Chandan Pal',  
  age: 19,  
  marks: 95,  
  city: 'Lucknow'  
}
```

11. DELETE ONE OPERATION

MongoDB Query:

db.students.deleteOne({roll:110})

Explanation: Deletes one document matching roll number 110.

A terminal window titled 'MONGOSH' with three colored window control buttons (red, yellow, green) in the top-left corner. The terminal displays a MongoDB command and its output. The command is entered on the first line, and the output is shown on the subsequent lines.

```
> db.students.deleteOne({roll:110})
{
  acknowledged: true,
  deletedCount: 1
}
```

12. DELETE MANY OPERATION

MongoDB Query:

```
db.students.deleteMany({  
  marks: {$lt:60}  
})
```

Explanation: Deletes all students having marks below 60.

A screenshot of a terminal window titled "MONGOSH". The terminal shows a MongoDB query being executed:

```
> db.students.deleteMany({  
  marks: {$lt:60}  
})  
{  
  acknowledged: true,  
  deletedCount: 2  
}
```

Removed roll numbers 111, 113 (marks below 60).

13. AND OPERATION — \$and

MongoDB Query:

```
db.students.find({
  $and:[
    {marks:{$gt:80}},
    {age:19}
  ]
})
```

Explanation: Finds students satisfying both conditions.

A screenshot of a terminal window titled "MONGOSH". The terminal shows a MongoDB query being executed:

```
> db.students.find({
  $and:[
    {marks:{$gt:80}},
    {age:19}
  ]
})
```

 The output shows two documents:

```
{
  _id: ObjectId('6aae9ea78f0f98eb52df2280'),
  roll: 101,
  name: 'Shrishti Singh',
  age: 19,
  marks: 85,
  city: 'Delhi'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df2283'),
  roll: 104,
  name: 'Anchal Gupta',
  age: 19,
  marks: 88,
  city: 'Varanasi'
}
```

Showing 2 of 10 matching documents.

14. OR OPERATION — \$or

MongoDB Query:

```
db.students.find({
  $or:[
    {city:"Lucknow"},
    {city:"Delhi"}
  ]
})
```

Explanation: Finds students from Lucknow or Delhi.

A screenshot of a terminal window titled "MONGOSH". The terminal shows a MongoDB query being executed: `> db.students.find({ $or:[ {city:"Lucknow"}, {city:"Delhi"} ] })`. The output displays two JSON documents. The first document is for a student with `_id: ObjectId('6aae9ea78f0f98eb52df2280')`, `roll: 101`, `name: 'Shrishti Singh'`, `age: 19`, `marks: 85`, and `city: 'Delhi'`. The second document is for a student with `_id: ObjectId('6aae9ea78f0f98eb52df2281')`, `roll: 102`, `name: 'Prathana Bajaj'`, `age: 18`, `marks: 72`, and `city: 'Lucknow'`.

```
MONGOSH
> db.students.find({
  $or:[
    {city:"Lucknow"},
    {city:"Delhi"}
  ]
})
{
  _id: ObjectId('6aae9ea78f0f98eb52df2280'),
  roll: 101,
  name: 'Shrishti Singh',
  age: 19,
  marks: 85,
  city: 'Delhi'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df2281'),
  roll: 102,
  name: 'Prathana Bajaj',
  age: 18,
  marks: 72,
  city: 'Lucknow'
}
```

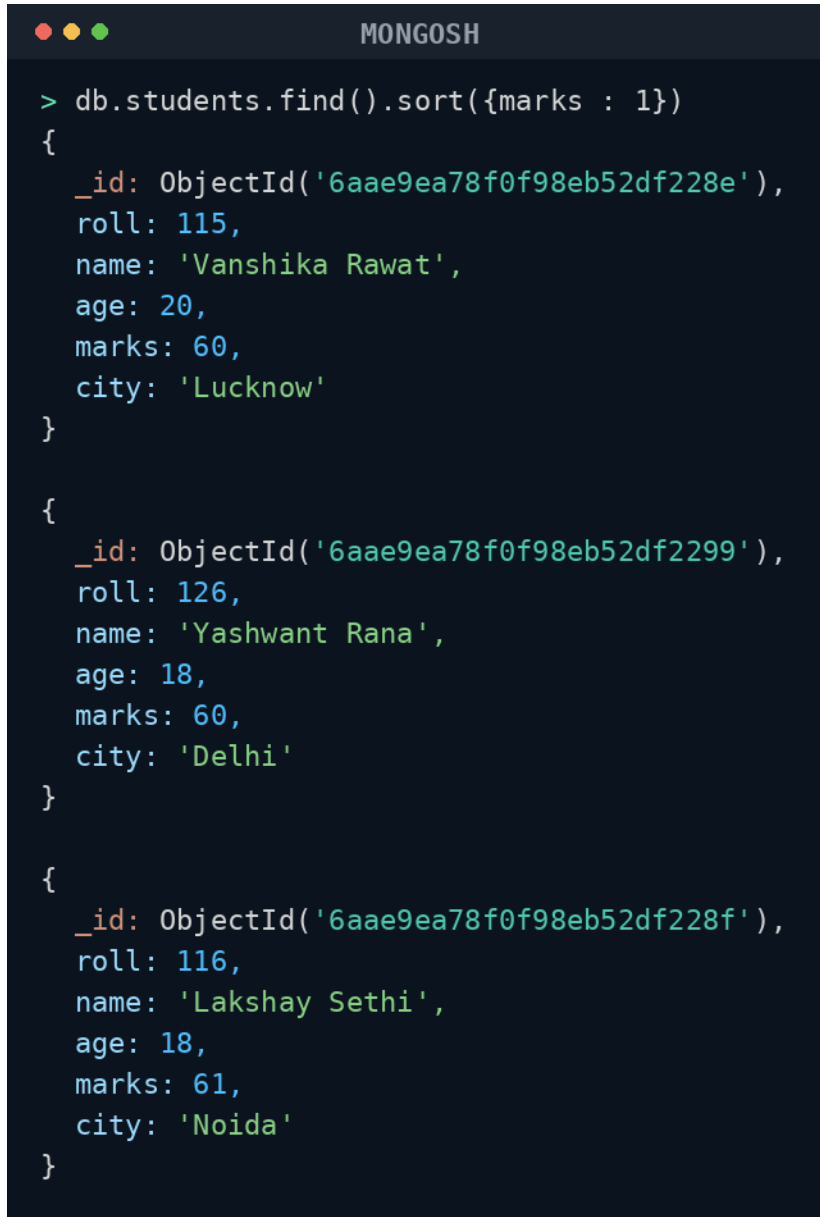
Showing 2 of 10 matching documents.

15. SORT OPERATION

A. MongoDB Query:

```
db.students.find().sort({marks : 1})
```

Explanation: Sorts students by marks from lowest to highest. Use 1 for ascending order.

A screenshot of a MongoDB terminal window titled 'MONGOSH'. The terminal shows a command to sort students by marks in ascending order. The output displays three student records, each as a JSON object. The first record is for Vanshika Rawat with a roll number of 115 and marks of 60. The second record is for Yashwant Rana with a roll number of 126 and marks of 60. The third record is for Lakshay Sethi with a roll number of 116 and marks of 61. The records are sorted by marks, and since the marks are the same for the first two, they are ordered by their _id values.

```
> db.students.find().sort({marks : 1})
{
  _id: ObjectId('6aae9ea78f0f98eb52df228e'),
  roll: 115,
  name: 'Vanshika Rawat',
  age: 20,
  marks: 60,
  city: 'Lucknow'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df2299'),
  roll: 126,
  name: 'Yashwant Rana',
  age: 18,
  marks: 60,
  city: 'Delhi'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df228f'),
  roll: 116,
  name: 'Lakshay Sethi',
  age: 18,
  marks: 61,
  city: 'Noida'
}
```

B. MongoDB Query:

```
db.students.find().sort({marks: -1})
```

Explanation: Sorts students by marks from highest to lowest. Use -1 for descending order.

```
MONGOSH

> db.students.find().sort({marks: -1})
{
  _id: ObjectId('6aae9ea78f0f98eb52df2284'),
  roll: 105,
  name: 'Chandan Pal',
  age: 19,
  marks: 95,
  city: 'Lucknow'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df228b'),
  roll: 112,
  name: 'Rohan Tripathi',
  age: 20,
  marks: 95,
  city: 'Prayagraj'
}

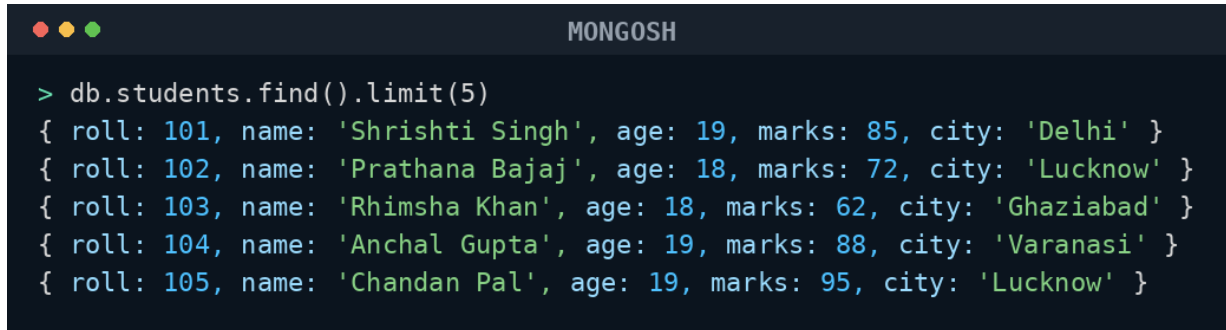
{
  _id: ObjectId('6aae9ea78f0f98eb52df2294'),
  roll: 121,
  name: 'Alisha Khan',
  age: 20,
  marks: 95,
  city: 'Noida'
}
```

16. LIMIT OPERATION

MongoDB Query:

`db.students.find().limit(5)`

Explanation: Displays only the first five documents.

A screenshot of a terminal window titled 'MONGOSH'. The terminal shows a MongoDB query and its output. The query is '> db.students.find().limit(5)'. The output consists of five JSON documents, each representing a student with fields for roll number, name, age, marks, and city. The documents are: { roll: 101, name: 'Shrishti Singh', age: 19, marks: 85, city: 'Delhi' }, { roll: 102, name: 'Prathana Bajaj', age: 18, marks: 72, city: 'Lucknow' }, { roll: 103, name: 'Rhimsha Khan', age: 18, marks: 62, city: 'Ghaziabad' }, { roll: 104, name: 'Anchal Gupta', age: 19, marks: 88, city: 'Varanasi' }, and { roll: 105, name: 'Chandan Pal', age: 19, marks: 95, city: 'Lucknow' }.

```
> db.students.find().limit(5)
{ roll: 101, name: 'Shrishti Singh', age: 19, marks: 85, city: 'Delhi' }
{ roll: 102, name: 'Prathana Bajaj', age: 18, marks: 72, city: 'Lucknow' }
{ roll: 103, name: 'Rhimsha Khan', age: 18, marks: 62, city: 'Ghaziabad' }
{ roll: 104, name: 'Anchal Gupta', age: 19, marks: 88, city: 'Varanasi' }
{ roll: 105, name: 'Chandan Pal', age: 19, marks: 95, city: 'Lucknow' }
```

17. SEARCH BY CITY

MongoDB Query:

```
db.students.find({city:"Lucknow"})
```

Explanation: Finds all students whose city is Lucknow.

A screenshot of a terminal window titled "MONGOSH". The terminal shows a MongoDB query being executed: `> db.students.find({city:"Lucknow"})`. The output displays three JSON documents. Each document has an `_id` field (an ObjectId), `roll`, `name`, `age`, `marks`, and `city` (all set to "Lucknow"). The first document is for Prathana Bajaj (roll 102, age 18, marks 72). The second is for Chandan Pal (roll 105, age 19, marks 95). The third is for Vanshika Rawat (roll 115, age 20, marks 60).

```
> db.students.find({city:"Lucknow"})
{
  _id: ObjectId('6aae9ea78f0f98eb52df2281'),
  roll: 102,
  name: 'Prathana Bajaj',
  age: 18,
  marks: 72,
  city: 'Lucknow'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df2284'),
  roll: 105,
  name: 'Chandan Pal',
  age: 19,
  marks: 95,
  city: 'Lucknow'
}

{
  _id: ObjectId('6aae9ea78f0f98eb52df228e'),
  roll: 115,
  name: 'Vanshika Rawat',
  age: 20,
  marks: 60,
  city: 'Lucknow'
}
```

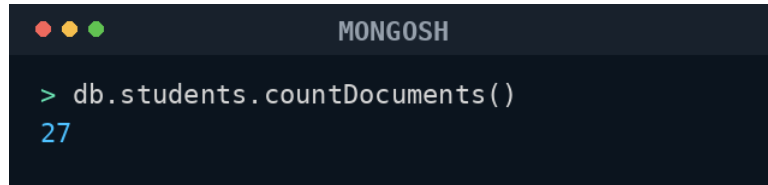
Showing 3 of 5 matching documents.

18. COUNT DOCUMENTS

MongoDB Query:

db.students.countDocuments()

Explanation: Counts documents currently present in the collection (after the delete operations).

A screenshot of a terminal window titled "MONGOSH". The window has a dark background with three colored window control buttons (red, yellow, green) in the top-left corner. The prompt ">" is followed by the command "db.students.countDocuments()". The output "27" is displayed below the command.

```
> db.students.countDocuments()  
27
```

MONGODB OPERATORS USED:

Operator	Meaning	Example
\$eq	Equal to	marks: {\$eq:88}
\$gt	Greater than	marks: {\$gt:85}
\$lt	Less than	marks: {\$lt:60}
\$gte	Greater than or equal to	marks: {\$gte:90}
\$lte	Less than or equal to	marks: {\$lte:70}
\$and	All conditions true	\$and:[...]
\$or	At least one condition true	\$or:[...]